SOP-AGILE SCRUM & KANBAN PROCESS

[CONFIDENTIAL]
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CHANGE HISTORY

This section describes the details of changes that have resulted in the current document.

SL. No.	Document Version No.	Change Description	Reason For change	Date Prepared By	Date Reviewed By	Date Approved By
01	0.1	Draft	Implementation of the Integrated Management System	Obul Prasad Reddy 21-Jul-23		
01	Issue B Version 0.1	Scrum Process development			Anitha N 21-Jul-23	
02	Issue B Version 0.2	Agile Scrum & Kanban Process	Agile methodology and Kanban framework along with Scrum and Kanban Flowchart's	Sachin Aramani 22-Jul-23	Santosh V 25-Jul-23	
03	Issue B Version 1.0	Agile Scrum & Kanban Process	Abbreviation, Responsibility, Scrum & Kanban Flow chart, Scrum & Kanban Process, Metrics	Sachin Aramani 26-Jul-23	Anitha N 27-Jul-23	Rama G 28-Jul-23
04	Issue B Version 1.1	Document formatting, font style to Poppins and alignment	Updated in line with the document guidelines	Devansh S 20-Dec-24	Anitha N 23-Dec-2024	Santosh V 31-Dec-2024
05	Issue C Version 1.0	Review of document	New version update of IMS	Anitha N 01-08-2025	Santosh V 08-Aug-25	Sanmangala MS 14-08-2025

1 Purpose

The purpose of this document is to define a procedure for agile methodology using Scrum and Kanban frameworks.

2 Scope

Applicable for all the projects which are following agile Scrum & Kanban methodology at Utthunga technologies.

3 Abbreviations

a. DOD: Definition of Done

b. PM: Project Manager

c. DM: Delivery Manager

d. PBI: Product Backlog Item

e. PBR: Product Backlog Refinement

f. PB: Product Backlog

4 Terms and Definitions

a. Product Owner: Is a person who oversees the development of product from start to finish and maximize their value.

b. Sprint: A set period during which specific tasks must be completed

c. User story: It is a small, self-contained unit of development work designed to accomplish a specific goal within a product.

d. Product Backlog: A prioritized list of work (user stories) for the development team that is derived from the roadmap and its requirements.

e. Sprint Planning: It is a process that breaks big projects into smaller parts common to the Scrum framework.

f. Sprint Backlog: is an ordered list of tasks that need to be completed in the duration of the current sprint.

g. Daily Scrum: is a 15-minute scrum ceremony where the Scrum Team collaborate and plan for the next 24 hours. It involves what was completed, any impediments, what is the next plan.

h. Sprint review: It is an informal meeting held at the end of a sprint, during which the team shows what was accomplished, while the stakeholders provide feedback. It's a collaborative working session rather than a one-sided presentation. Sprint Review is also called as Sprint Demo

i. Sprint Retrospective: It is a scrum ceremony where the Scrum Team inspect and adapt their working practices.

j. Product Backlog Refinement: It is an ongoing process in which the Product Owner and the Scrum Team collaborate to ensure that items on the Product Backlog

k. Kanban Board: It is an agile project management tool designed to help visualize work, limit work-in-progress, and maximize efficiency (or flow).

l. Kanban Card: It is a visual representation of a work item on a Kanban board.

m. Metric: It is a quantifiable measure that is used to track, compare, and assess performance or processes

n. Burn down chart: It is a graphic representation of how quickly the team is working through user stories.

o. Sprint Velocity: It is the measure of how much a team can realistically accomplish within a sprint cycle.

p. Cycle Time: It is a measure of how long a project takes from starting the work to completion.

q. Work In progress: It is the number of tasks or work items a team has on their schedule at any given time.

r. Sprint Cancelation: It is termination of sprint before it's intended completion date.

5 Responsibilities

5.1 Scrum Responsibility

a. Product owner shall Define and prioritize the Product Backlog & work items.

b. Scrum Master will create the Sprint Backlog

c. Team members such as developers, testers and others will help in creating Sprint Backlog

d. Scrum Master will take updates from team members in daily scrum.

e. Product Owner, Scrum Master, Team members & Business stakeholders will be part of Sprint review.

f. Product Owner, Scrum Master & Team will be part of sprint retrospective.

5.2 Kanban Responsibility

a. Product owner shall Define and prioritize the Product Backlog & work items.

b. Technical Lead shall Implement and maintain Kanban.

c. Product Owner Shall decide which Kanban card to be placed on Kanban board.

d. Kanban team shall select and pull the work from backlog.

6 RACI

Process	Responsible	Accountable	Consulted	Informed
Agile Scrum	Scrum Master	Product owner	Customer	Customer, Delivery Manager & Quality
Agile Kanban	Technical Lead	Product Owner	Customer	Customer, Delivery Manager & Quality

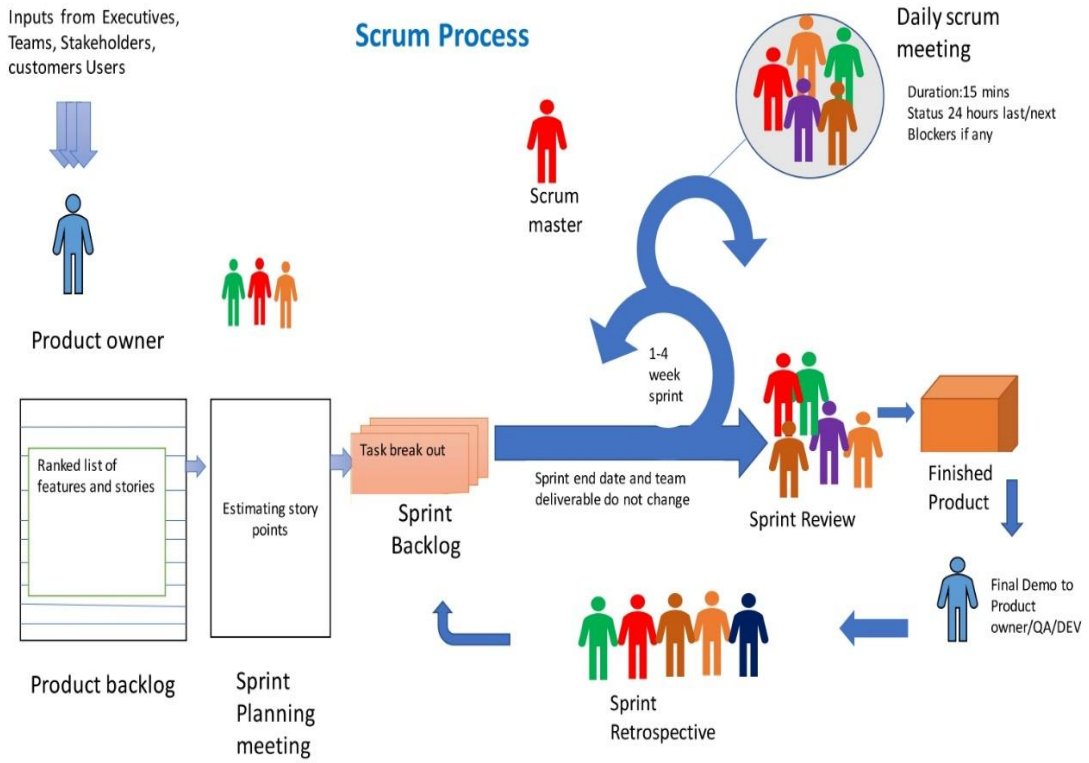
7 SIPOC

Supplier	Input	Process	Output	Customer
Customer	Requirement Specification document	Brainstorming	Product Backlog	Scrum Master

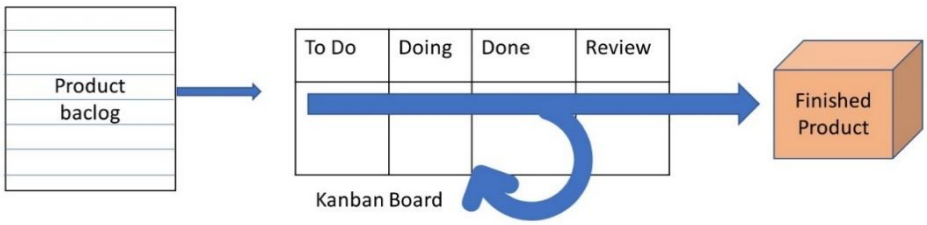
Product Owner	Backlog document	Sprint Planning Brainstorming and discussion with team	Sprint Backlog	Team
Scrum Master	Sprint Backlog	Daily Scrum Task execution Design, Code & Test case review. Sprint review	Developed Product	Customer
Supplier	**Input**	**Process**	**Output**	**Customer**
Customer	Requirement Specification document	Brainstorming Prioritizing work items	Product Backlog	Kanban Team
Product Owner	Product Backlog	Kanban board with categories and stages Kanban Card	Kanban board with Kanban Card	Team
Product Owner	Kanban Board with Kanban Card	Kanban Card Task execution Pull System Design, Code & Test case review.	Developed Product	Customer

8 Overall Process Flow Chart

Scrum Process Flow



Kanban Process Flow



9 Process Description

9.1 Scrum

a. Scrum Procedure

- The Product Owner will create the product backlog using requirement specification document and prioritize the work items.
- The Product Owner defines the acceptance criteria for each user story & definition of done for each sprint. Most of the time Definition of done will remain same for all sprints however there could be some changes if there is a change in the scope of the sprint.
- Product owner prioritizes the product backlog for at least 3 sprint and the prioritized item shall be added in the respective sprint backlog. PO takes input from customer for prioritizing the PBI.
- Sprint backlog will be shared with the client for their approval.
- The Scrum master conducts the Sprint Planning Meeting at the beginning of every sprint.
- At Sprint Planning meeting the Product owner along with Scrum team re-estimate the work items from backlog selected for that sprint and defines the sprint goal. Sprint planning should not go beyond 8 Hrs. for a 1-month sprint.
- The Scrum Master conducts Daily Scrum for not more than 15 minutes.
- The scrum team will join daily scrum and update the status of the tasks, impediments if any and plan for next 24 hours.
- The scrum team completes the planned tasks as per the acceptance criteria and DOD.
- The tasks which are not completed in the current sprint will be moved back to the product backlog.
- The Scrum team will conduct the sprint review along with Product Owner & Customer. The observations of the Demo will be fixed, and release will be planned.
- Sprint Retrospective meeting will be conducted at the end of the sprint with Scrum team by analyzing current sprint with best practices, lesson learnt, impediments and adapting the best practices, lesson learnt for subsequent sprint.

- Product backlog refinement is done to re-prioritize & re-estimate the backlog items.

9.2 Kanban

b. Kanban Procedure

a. The Product owner will create the Product backlog using Requirement specification document and prioritize the work items.

b. The Technical lead will create the Kanban board by categorizing all stages of work.

c. The stages to be listed side to side like in queue, in progress, review & done.

d. The Product owner will create the Kanban cards with title, description, acceptance criteria and assign the tasks to Kanban Team

e. The Kanban team will use pull system to move the Kanban cards as per the work.

f. The Kanban team moves the cards to done or completed stage once the work item is completed and review comments are closed and approved.

10 Information Security requirements

a. The information security related requirements shall be included in the Product backlog for new information systems or enhancements to existing information systems.

b. Secure access (user registration and deregistration) to various applications, Project portals

c. Ensure data privacy, confidentiality and IP requirements are clearly specific in the contract.

d. Information Security requirements in the contract are clearly put into place.

e. Compliance requirements based on different geographies are adhered to in projects.

f. Confidentiality of information and project details is maintained by the team.

11 Metrics

a. Scrum Metrics

Following are some of the recommended metrics however, the team can refer the other metrics from Metrics-Guidelines: UT/IM/MAI/GDL/01

Metric	Formula
Velocity	No of work item or story points/sprint span in days
Sprint Burn Down	It is a graph that plotted using story points or effort left to do to that of sprint duration
Epic & Release Burn down	It is a graph that plotted using total epic or effort to that of planned epic duration

b. Kanban Metrics

Following are some of the recommended metrics however, the team can refer the other metrics from Metrics-Guidelines: UT/IM/MAI/GDL/01

Metric	Formula
Cycle time	Analysis start date + Design + Coding + Testing done
Process efficiency	On-hold time - cycle time/cycle time
Work in Progress	No of story points in progress
Work item age	(Current time unit – start time unit) + 1

12 References

a. SOP Kanban for agile development.

b. SOP Scrum for agile development.

c. Metrics-Guidelines

13 Annexures

NA

14 Forms

a. Project Execution Agile UTPMPPPTMP14

b. Project Plan Template scrum UTPMPPPTMP10

c. Kanban Board UTPMPPPTMP13